Data Mining I
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Exercise 3

1. Describe a potential problem of using **Information Gain** as a splitting criterion in decision trees. Explain why the **Gain Ratio** measure is a way to overcome this problem.

2. Consider the training examples shown in Table 1 for a binary classification problem. Calculate **Information Gain** and **Gain Ratio** for each of the attributes *ID*, *Gender*, *Graduation* and *Eye Color*. Use multiway splits when possible.

3. There exist several approaches to measure the generalization error rate of a decision tree T. The simplest measure is termed as **optimistic error estimate** $\mathrm{err}_o(T)$, which is simply the error rate of T on the training set. In contrast, the **pessimistic error estimate** $\mathrm{err}_p(T)$ also incorporates the "complexity" of T, quantified by the number of leaf nodes.

   Let k be the number of leaf nodes and N_{train} be the number of training instances. The pessimistic error estimate is computed as:

$$\mathrm{err}_p(T) = \mathrm{err}_o(T) + \Omega \cdot \frac{k}{N_{\text{train}}}$$

   where Ω is a hyperparameter that makes a trade-off between minimizing training error and minimizing model complexity.

   A third generalization error estimate is **reduced error pruning** ($\mathrm{err}_{\text{rep}}$), which is the error rate of T on a validation set.

   Calculate $\mathrm{err}_o(T)$, $\mathrm{err}_p(T)$, and $\mathrm{err}_{\text{rep}}(T)$ for each of the two decision trees in Figure 1.

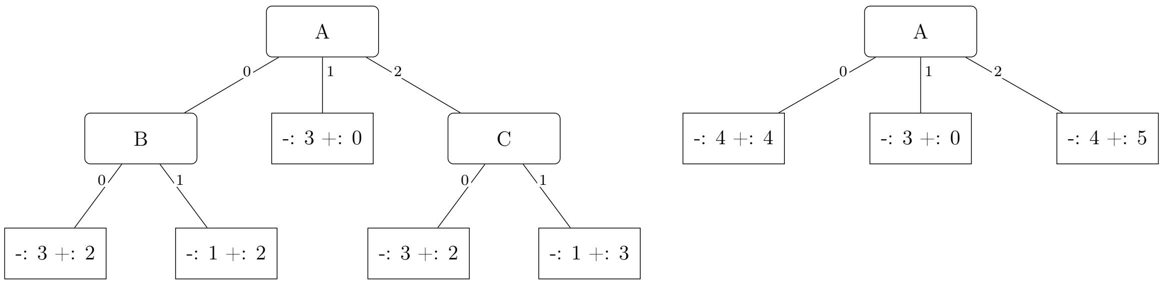
   Assume that each leaf node is labeled according to the majority class of training instances that reach the node. When calculating the pessimistic error estimate, assume that $\Omega = 2$.

   Reduced error pruning steps:

   - Consider each node for pruning.
   - *Pruning* = removing the subtree at that node, make it a leaf and assign the most common class at that node.
   - A node is removed if the resulting tree performs no worse than the original on the validation set — removes coincidences and errors.
   - Nodes are removed iteratively choosing the node whose removal most increases the decision tree accuracy on the graph.

Table 1: Dataset for task 2.

ID	Gender	Graduation	Eye Color	Class
1	F	College	brown	1
2	F	College	brown	1
3	F	College	gray	0
4	F	High School	blue	0
5	F	High School	blue	0
6	F	High School	brown	0
7	F	High School	gray	1
8	F	Middle School	blue	1
9	F	Middle School	blue	1
10	F	Middle School	brown	1
11	M	High School	blue	0
12	M	High School	brown	0
13	M	High School	brown	0
14	M	High School	gray	0
15	M	High School	green	0
16	M	High School	green	0
17	M	Middle School	brown	1
18	M	Middle School	gray	1
19	M	Middle School	green	1
20	M	Middle School	green	1



ID	A	B	C	Class
21	0	1	0	+
22	0	1	1	+
23	1	0	0	-
24	1	1	1	+
25	2	1	0	+

Figure 1: Two decision trees generated from the same training set and validation data.